

ADARSH GUPTA

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EDUCATION

Indian Institute of Technology, Dharwad

2022 – 2026

B.Tech. in Mathematics and Computing | CPI: 8.88 / 10.0

Relevant Coursework: Advanced Optimization Theory, Linear Algebra, Probability & Random Processes, Data Structures & Algorithms, Distributed Systems, Database Management Systems

EXPERIENCE

Arctan Technologies

Jan 2026 – Present

Data Science & Speech AI Intern

Bangalore, India

- Accelerated conversational agent training loops by 60% as measured by reduced onboarding cycles, by designing an end-to-end Voice Roleplay Bot integrating naturalized dialogue management and automated speech scoring systems.
- Expanded baseline model training datasets by 100+ parallel audio hours across distributed cluster nodes, by architecting a high-throughput text-to-speech (TTS) synthesis and Dynamic Time Warping (DTW) alignment pipeline.
- Improved localized accent representation for Indian speech dialects by 15% against baseline architectures, by developing a cross-lingual latent-space mapping framework using S3 tokens in CosyVoice and fine-tuning acoustic diffusion layers.

Dvara E-Registry

Jun 2025 – Dec 2025

Data Science / ML Engineering Intern

Remote

- Outperformed baseline algorithmic models by 25% by dropping 14-day prediction errors to 10% MAPE (and 4-week errors to 14% MAPE), through engineering a hybrid LightGBM + ARIMA/SARIMA framework optimized with 365-day daily seasonality and PACF features.
- Minimized regional price prediction variance across large-scale datasets by 22% across multiple test markets, by constructing multi-tiered, district-level forecasting architectures utilizing deep LSTM networks and robust outlier handling.
- Optimized corporate procurement strategies across 15 agricultural districts by generating a 15% increase in data-backed decision efficiency, through translating complex statistical modeling insights and volatility trade-offs to stakeholders.

PROJECTS

Flood Detection – Computer Vision Classifier

Aug 2023 – Sept 2023

TensorFlow · Keras · Sentinel-1 SAR Data · Remote Sensing

- Achieved 90% regional classification accuracy on the Flood11 benchmark dataset by constructing a 5-layer Convolutional Neural Network (CNN) optimized with Sentinel-1 SAR remote sensing data, batch normalization, and dropout regularization.
- Enhanced model generalization and minimized overfitting trends by 12% across structural evaluations, by integrating spatial data augmentation strategies and analyzing performance metrics via multi-class confusion matrices.

Cricket Fantasy Analytics & Interpretable NLP Pipeline

Nov 2024 – Jan 2025

Python · XGBoost · LSTM · Natural Language Processing (NLP/NLU) · SHAP

- Identified 6.86 out of 11 top-performing athletic targets on average against real-world game statistics, by architecting an automated multi-source data engineering pipeline applying nonlinear optimization techniques via XGBoost and CatBoost models.
- Elevated model decision transparency and end-user engagement by 35% using post-hoc SHAP explainability matrices, by deploying a LLaMA-driven Natural Language Understanding (NLU) generation layer to interpret complex outputs.

Distributed Peer-to-Peer Audio Streaming Platform

Mar 2024 – Apr 2024

Python · Sockets · Network Topology · PyAudio · Multithreading

- Architected concurrent audio streaming for up to 8 real-time nodes under network constraints, by building a decentralized peer-to-peer communication topology using thread-safe, fault-tolerant UDP/TCP signaling architectures.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, Java, JavaScript, TypeScript

AI/ML Core: Deep Learning, Automatic Speech Recognition (ASR) Fundamentals, Text-to-Speech (TTS), Natural Language Processing & Understanding (NLP/NLU), Computer Vision (CNNs), Reinforcement Learning (RL) Foundations

Frameworks & Tools: PyTorch, TensorFlow, Scikit-learn, Transformers, OpenAI/LLaMA APIs, SHAP, Streamlit, Git

Systems & Architecture: Distributed Systems, Apache Spark / Hadoop, Docker, Nginx, Linux, Bash, FastAPI, WebSocket

Certifications: [Machine Learning Specialization \(Stanford / DeepLearning.AI\)](#), [Introduction to Data Science \(Cisco\)](#), [SQL Advanced \(HackerRank\)](#), [InterIIT Tech Meet 13.0 \(IIT Bombay\)](#)

LEADERSHIP

- Led an IIT Dharwad engineering team of 12 peers during a 6-week technical sprint at the Inter-IIT Tech Meet 13.0 (IIT Bombay), managing ambiguous deliverables and presenting technical architectural trade-offs to judging panels.